Tire Coast Down Noise Evaluation Procedure

1 Scope

Note: Nothing in the standard supersedes applicable laws and regulations unless specific exemption has been obtained.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. Determination of tire rolling noise.

1.2 Foreword. Due to the reduction of legal pass by noise limits (e.g., 74 dB (A) with introduction of directive 96/20EEC) tire noise becomes important as a major noise source.

Above test procedure basically corresponds to directive 92/23/EEC, however, it determines tire noise at a speed of 55 mph (mean speed according to ISO 362) and not at 80 km/h.

1.3 Applicability. For all passenger car tires.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

92/23/EEC	ISO 362
96/20/EEC	ISO 10844

2.2 GM Standards/Specifications.

None

2.3 Additional References. None.

3 Resources

3.1 Facilities. Test track according to ISO 362 with a pavement as specified in ISO 10844.

3.2 Equipment. Sound level meter or other noise measuring system according to ISO 362.

3.3 Test Vehicle/Test Piece. Vehicle fitted with 4 new tires of same make, however, test tires should have undergone "break-in" ride (to remove any mold release agents on tread surface).

3.4 Test Time. Not applicable.

3.5 Test Required Information. Not applicable.

3.6 Personnel/Skills. Not applicable.

4 Procedure

4.1 Preparation. Measurement set up according to ISO 362 (see Appendix A).

4.2 Conditions.

4.2.1 Environmental Conditions. The ambient influences (air temperature, etc.) shall be maintained during test runs, if possible.

The following applies according to ISO 362:

- Ambient temperature: 0 to +40°C.
- Wind speed: < 5 m/s
- Ambient noise level: > 10 dB(A) below vehicle noise level

The test track shall be dry and free of sound affecting obstacles (see ISO 362).

4.2.2 Test Conditions. Deviations from the requirements of the procedure shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.2.2.1 The tire pressure (cold) shall be set to the nominal value.

4.2.2.2 The measurements shall be performed with curb weight (= operational condition) including fuel and operating material, vehicle equipment and driver.

4.2.2.3 The test vehicle incl. engine, tires etc. shall have operating temperature prior to each test.

4.3 Instructions. Several coast-down measurements shall be performed at different speeds. The vehicle shall be accelerated, and prior to reaching line AA', the transmission shall be put into neutral position and the engine shall be stopped **(Warning! Steering lock shall not be engaged).**

The test speeds shall be determined at vehicle position 0 m (= position of microphone) and shall be in a range of 50 to 60 km/h.

The tire rolling noise levels at left and RH-side shall be measured in a distance of 7.5 m to the vehicle center line.

Following data shall be documented additionally to noise values:

4.3.1. Tire data (supplier, size, pattern designation, DOT No., tire pressure, tread-depth).

4.3.2. Environmental conditions, e.g. air temperature, test track temperature, air pressure, air humidity, wind velocity, environmental noise.

4.3.3 Applied measuring system and test track.

5 Data

5.1 Calculations.

5.1.1 Graphic Evaluation. The measured noise level values are plotted versus car speed. A linear regression line is drawn through these points.

The sound pressure value of the linear regression at the speed of 55 km/h then shall be determined and this test result will be recorded.

Note: Although there is an exponential relation between sound pressure and speed, a linear interpretation is acceptable, since above procedure refers to only a small speed range.

5.1.2 Mathematical Evaluation. The rolling noise test value at a reference speed of 55 km/h may also be determined from the measured values by means of a mathematical regression analysis.

5.2 Interpretation of Results.

5.2.1 Temperature Correction. Since tire rolling noise is temperature-dependent and measurement might be performed at different temperatures, the results determined according to Paragraph 5.1 shall be corrected to a special reference temperature by an empirically determined coefficient.

$$L_R = L_{Mess.} + K_T\,(T_{Mess.} - T_{Ref.})$$

L_R = Rolling noise at reference temperature

L_{Mess} = Determined rolling noise

K_T = Coefficient of correction

T_{Mess} = Measurement temperature

T_{Ref} = Reference temperature

5.2.2 Limit. Depending on the legal target, the rolling noise may be limited (in-house) to obtain appropriate and leveled noise source rates for low pass-by noise results.

5.2.3 Rules. Following determinations are valid at to fulfill EC directive 96/20/EEC at present:

Rolling noise limit:

$L_R \leq 69$ dB (A)

$T_{Ref.} = +10°C$

$K_T = 0.04$ dB (A)/°C

5.3 Test Documentation. With reference to this test procedure, the results of the measurements shall be presented in a test report.

6 Safety

This procedure may involve hazardous materials, operations, and equipment. This method does not propose to address all the safety problems associated with its use. It is the responsibility of the user of the method to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary. None.

7.2 Acronyms, Abbreviations, and Symbols. None.

8 Coding System

This test procedure shall be referenced in other documents, drawings, VTS, CTS, etc. as follows:

Test to GMW14469.

9 Release and Revisions

9.1 Release. This standard originated in September 2005 replacing GME L/R-10B-3. It was first approved by the Global N&V Acoustic and Airborne Noise IDSR Harmonization Team in December 2005. It was first published in December 2005.

9.2 Revisions.

Rev	Approval Date	Description (Organization)
A	OCT 2006	Paragraphs 5.2.1 and 5.2.2 revised. (Global N&V Acoustic and Airborne Noise IDSR Harmonization Team)

Appendix A

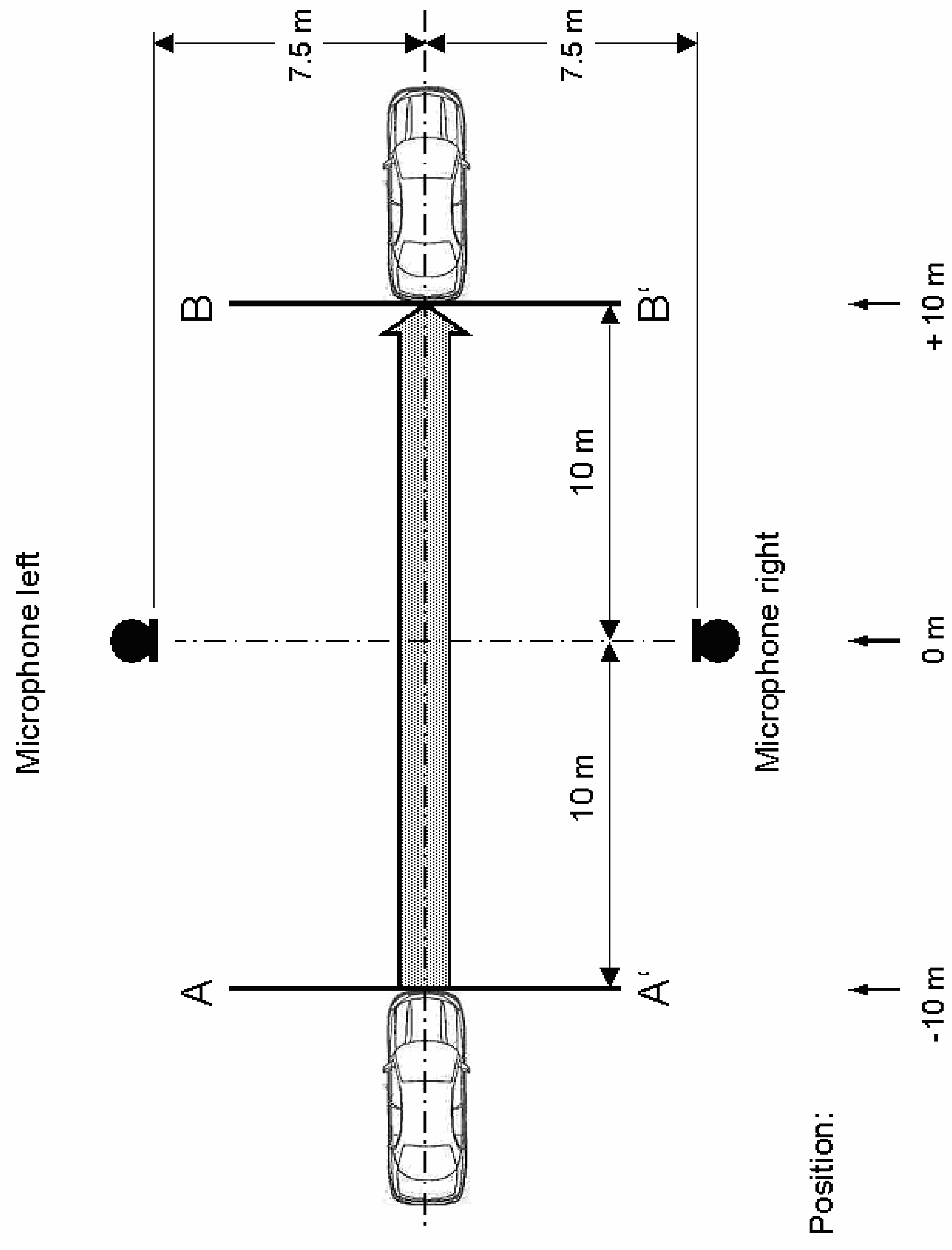

Figure A1: Dimensions of Test Track According to ISO 362